



The role of plant species on runoff and soil erosion in a Mediterranean shrubland

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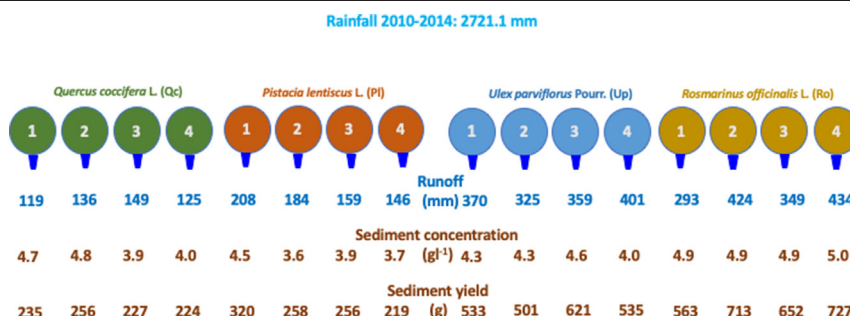
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HIGHLIGHTS

- Mediterranean shrubs plant species protect the soil from erosion.
- Plant sprouters generate 2.4 times less runoff than the plant seeders.
- Plant seeders induce higher ($\times 2.8$) soil and water losses than plant sprouters.
- 10 rainfall events generated 92% of the soil erosion.
- Litter was found as a key factor to reduce the soil losses.

GRAPHICAL ABSTRACT



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ABSTRACT

Shrubland is a Mediterranean biome characterized by densely growing evergreen shrubs adapted to fire events. To date, scientific research has focused on the impact of vegetation on soil erosion mainly through the control that plant biomass or plant cover exerts on sediment delivery and runoff discharge, being the individual plant species influence on hydrological and erosional processes not achieved in detail. The objective of this research is to determine: i) runoff and soil losses in a shrubland-covered rangeland at Sierra de Enguera, Spain; and ii) how four plant species affect soil and water losses. We measured soil cover, soil properties, runoff discharge and sediment yield under natural rainfall for five years (2010–2014) in a typical shrubland burnt in 1999. Four plant species were selected with 4 plots each: *Ulex parviflorus* Pourr., *Pistacia lentiscus* L., *Quercus coccifera* L. and *Rosmarinus officinalis* L. Despite that the soil properties and plant cover did not exhibit statistically significant differences among plant species, the runoff discharge was lower on *Q. coccifera* (4.87%, SE 0.24) and *P. lentiscus* (6.24%, SE 0.51) than on *U. parviflorus* (13.41%, SE 0.58) and *R. officinalis* (13.84%, SE 1.23). Sediment concentrations were, respectively, 3.91, 4.33, 4.31 and 4.88 g l⁻¹, and the differences between *R. officinalis* and the other species were statistically significant. The runoff discharge determined differences in soil erosion rates among the plant species with lower rates on *P. lentiscus* (1.36 Mg ha⁻¹ y⁻¹) and *Q. coccifera* (1.53 Mg ha⁻¹ y⁻¹), than on *U. parviflorus* (3.17 Mg ha⁻¹ y⁻¹) and *R. officinalis* (3.85 Mg ha⁻¹ y⁻¹). This long term in situ study indicated that *Q. coccifera* and *P. lentiscus* are more efficient in controlling runoff discharge and soil losses than *U. parviflorus*.

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and *R. officinalis* one decade after a fire. We discuss these results in light of the recent findings by the scientific community of the role of the canopy cover (rainfall interception), soil macropore and root system, and the water repellency that control the hydrological response of the soil (e.g. runoff generation, infiltration). The information supplied by 5 years of research is relevant for restoration and rehabilitation programs and advise that *Q. coccifera* and *P. lentiscus* are the most efficient plant species to control soil and water losses within the Mediterranean shrubland. This is an applied science approach for a better management of rangelands.

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1. Introduction

Plants influence hydrological and erosional processes. Depending on the plant species, evaporation (Schreiber and Riederer, 1996), rainfall interception (Gardon et al., 2020), stemflow (Yang and Wei, 2019), throughfall (Magliano et al., 2019), moisture (Wang et al., 2019), soil infiltration (Bui et al., 2019), watershed runoff discharge (Keesstra et al., 2019) and soil losses (Nouwakpo et al., 2020) are modified. Changes in vegetation result in changes biogeochemical cycles (Shevliakova et al., 2009; Piao et al., 2005). This is relevant in dynamic ecosystems where changes in vegetation are registered across time (Williams et al., 2004) and space (Mu et al., 2012). This is why areas affected by fires are landscape laboratories due to their dynamic behavior, from the point of view of plants and soil, and their influence on runoff generation and soil erosion along the recovery period (Rhoades et al., 2019). This is relevant under the climate change and to face the climate change impact on rangelands (García-Ruiz et al., 2010; López-Vicente et al., 2021).

Mediterranean-type ecosystems are affected by recurrent fires (Pausas et al., 2009) and, as in other biomes, fire is a natural process, which is strongly influenced by human use of the land (Keeley et al., 2005; Pyne, 2019). The impact of each fire event results in the partial removal of the vegetation cover and changes to soil physico-chemical properties. These changes lead to alterations in water flow paths and water retention and increases in soil losses (Hu et al., 2020; Lestienne et al., 2020; Wang and Fu, 2020). This period with higher soil erosion rates as a consequence of fire is known as the “window of disturbance” (Prosser and Williams, 1998). Loss of vegetation is a key factor that controls the accelerated soil erosion rates triggered after fires. As vegetation regrows, soil erosion and infiltration rates tend to return to pre-fire values (Plaza-Álvarez et al., 2019; Arellano-Pérez et al., 2020; Ebel, 2020; Nouwakpo et al., 2020).

Plant species recover after fires at different rates and the mix of plant species changes during the recovery period (Keeley et al., 2005). This regrowth and succession of plant species exert control on the return to pre-fire conditions (Tepley et al., 2018). However, there are few studies that examine the effect of individual plant species on soil and water losses once the vegetation cover is reestablished, even though it has been documented that plants control runoff generation and sediment production (Chau and Chu, 2017; Vannoppen et al., 2017; Kervroëdan et al., 2018; Kervroëdan et al., 2019). Most of the research on the relationship between soil erosion and vegetation has focused on total cover or biomass (Diaz-Delgado and Pons, 2001; Fernández et al., 2019; Leunda et al., 2020) and these results have been incorporated into older soil erosion models (Nearing, 2000; Jetten et al., 2003) and more recent models (Aksoy et al., 2020; Jarrah et al., 2020). To further our scientific understanding of post-fire recovery and provide insights into restoration strategies, it is relevant to study how individual plant species will affect soil and water losses. Previous work carried out in America and Europe was mainly developed in California and the south of Europe due to the fact that forest fire is more recurrent in Mediterranean type ecosystems.

To date, few studies have focused on individual plant species and type of plant during research on runoff generation and soil erosion particularly in Mediterranean ecosystems where vegetation recovery is very dynamic (Pausas et al., 1999; Keeley et al., 2005; Lucas-Borja et al., 2018). Some studies have investigated the role of individual plants

on soil erosion under laboratory conditions or controlled field experiments using rainfall simulators (Cerdà and Doerr, 2005). Berendse et al. (2015) investigated how the loss of plant species reduces soil erosion resistance, and De Baets et al. (2007) reported on the importance of root characteristics of Mediterranean plant species to reduce concentrated runoff. Guerrero-Campo and Montserrat-Martí (2000) were the pioneers in research on how floristic composition influences soil erosion rates. Chen et al. (2004) previously found that species-diversified plant covers enhanced resistance to soil erosion. Xu et al. (2009) studied how plant species-controlled soil erosion rates in Southwest China. Earlier research by Tsuyuzaki (1990) found a good correlation between soil erosion and species composition at a ski area in Hokkaido. Most of the research has compared different plant communities with soil erosion rates. In general, long-term measurements that will shed light on how the vegetation affects hydrological and erosional processes in Mediterranean ecosystems (Francos et al., 2016; Úbeda et al., 2019) will be very beneficial to help plan more efficient afforestation strategies, and to develop more accurate soil erosion models.

Afforestation aims to reestablish the vegetation cover, and restore the ecosystem by improving soil organic carbon, soil fertility and vegetation diversity (Andrés and Ojeda, 2002; Fernández-Ondoño et al., 2010; Laganier et al., 2010). It is well known that the plant species selected for restoration will determine species richness and the fate of recovery and thus the ecosystem evolution (Amici et al., 2012). Moreover, afforestation has other objectives in addition to forest restoration. Afforestation projects are also designed to restore vegetation cover with the purpose of reducing soil and water losses (Zhang et al., 2008; Wang et al., 2014). However, in this context little information exists about the effect of individual plant species on soil and water losses, and this is one of the topics emphasized in this research. Under natural recovery it has been documented different plants recover faster than others (Úbeda et al., 2006) but there is no information about how these plant species affect hydrological and erosional processes. Furthermore, there is also a need to know how rainfall controls soil properties, soil and water losses, and hence the vegetation recovery (Francos et al., 2019).

The plant species selected for restoration will determine species richness and the fate of recovery and thus ecosystem evolution through the control they exert on soil and water losses, as they will determine the soil quality and ecosystem health (Verity and Anderson, 1990; Li et al., 2000). Research results on the impact of vegetation recovery on soil quality are abundant, especially studies on fire-affected land and projects where rehabilitation, afforestation and restoration activities were carried out (Faucette et al., 2006; van Hall et al., 2017). Plant species are key factors that determine the recovery after a fire due to the control they exert in the soil water repellency degree (Plaza-Álvarez et al., 2018). The fate of the post-fire is upon the fire characteristics and the plant recovery as both determine the water repellency in Mediterranean ecosystems (Fernández-García et al., 2021; Zema et al., 2021). Given the paucity of research about effect of individual plant species on soil and water losses within the scientific community, this research strives to fill this knowledge gap by presenting data on how four different plant species from the Mediterranean shrubland affect soil and water losses throughout a five-year study period in an area affected by fires. This investigation was inspired by the research developed by Cerdà and Doerr (2005) that explored the relationships of plant species with soil erosion during a rainfall simulation experiment (55 mm of rainfall in 1 h). Rainfall simulation

experiments give information about the plant and soil response under a fixed rainfall intensity. Here we monitored 16 plots with 4 different species under natural rainfall events for 5 years. This will complement previous findings that determined how plants from the shrubland ecosystems affect soil and water resources.

The aim of this paper is to assess the influence of different plant species on runoff and soil erosion in a typical Mediterranean shrubland. We hypothesize that each plant type has different hydrological behavior, which is relevant to understand the role of shrubland in the runoff generation and soil loss, and then, to plan rehabilitation and restoration programs.

2. Material and methods

2.1. Study area

We selected Sierra de Enguera as a typical Mediterranean shrubland ecosystem where fire is recurrent. The experimental soil erosion station at Sierra de Enguera is a rangeland located inland in the Valencia province (60 km from the Mediterranean coast), Eastern Iberian Peninsula, at an altitude of 765 m.a.s.l. (38° 55' N, 00° 49' W). This land was used as a pastureland by goats and sheep until the 1960s and for biomass production, however, due to the migration of the population of the mountains in the Mediterranean to the coastal and urban areas the abandonment took place during the 20th century. The vegetation recovery resulted in the growth of the shrubland and Aleppo pine (*Pinus halepensis* Mill.) after afforestation and natural recovery. The original vegetation of the Sierra de Enguera was dominated by *Quercus ilex* subsp. *rotundifolia* L. before the arrival of the human exploitation 3000 years ago that opened the forest and developed a savanna-like landscape similar to the Dehesa in Western Spain and Montados in Portugal. During the 19th century, the railway expansion in Spain transformed the oak forest into a pine plantation to produce railway sleepers. Fire was always a key process in the human transformation of the land as prescribed fires were used as a tool. However, the land abandonment along the 1960s resulted in more common and widespread wildfires. In 1979 a fire burnt the Macizo del Caroig and the Sierra de Enguera for four days (July 19–22, 1979) and burned 44,000 ha. After the recovery, fires became common and recurrent until now and affect mainly the shrublands and the Aleppo pine covered areas. The second and most recent wildfire recorded at the study site took place in 1999 (August 15) when 3196 ha burned.

Climate is typical Mediterranean (Csa: Köppen) with a mean annual temperature of 12.7 °C (Cerdà et al., 2017), with January as the coldest month (9.8 °C) and August the warmest (25.7 °C). Mean annual rainfall is 540 mm with a three-month recurrent Mediterranean drought (June 15th until September 15th). Although in the study area the total amount of rainfall is low, there are intense rainfall events that reach amounts higher than 100 mm day⁻¹. The landforms are characterized by a succession of plateaus and deeply incised valleys. This is a consequence of the weathering (dissolution) of the Cretaceous carbonate rocks which are the main parent material at the study area and in Eastern Spain. The experimental setup was installed on a hill on limestone. Rock fragments and rock outcrops are present along the upper slope. Shrubs covered 70% of the ground and they use to be located in patches of 3 to 6 m in diameter with a mixture of different species. The distance between shrubs patches ranged from 0.3 to 1 m. Rock outcrops did not affect the experiment that was located in the middle of the slope. The soils are classified as Inceptisols (Soil Survey Staff, 2014). The vegetation is composed of a mixture of Mediterranean shrubland species where the main shrubs are *Ulex parviflorus* Pourr., *Pistacia lentiscus* L., *Erica multiflora* L., *Juniperus oxycedrus* Sibth. & Sm., *Quercus coccifera* L. and *Rosmarinus officinalis* L.

2.2. Experimental design and methodology

The experimental installation was established from January until March 2010 and the measurements were started on October 1st 2010,

11 years after the last fire, and when the vegetation cover was already established. The sixteen plots were selected after a survey in the study area (12 ha) to select representative plants individuals, and also to select plots that allow to install plots and collectors. This was not always possible due to the rugged terrain. We selected 40 sites in the Sierra de Enguera, and after a characterization of the plant species and the shrub we selected the study site as representative one.

To avoid the impact of plot construction on the sediment and water delivery the first nine months of overland flow and sediment data are not included in this research to avoid artifacts due to the high delivery of sediments immediately after the plot installation. Usually, after the installation of the plots there are more sediment available in the soil surface, and the soil losses are higher due to this artifact. The measurements took place from October 1st 2010 until September 30th 2014. Sixteen circular plots were constructed with borders fabricated from aluminium sheets (1 mm width x 60 mm height) installed in the soil with a rubber hammer (1 cm depth) under the shrubs. The plots were built in a circular shape to better adapt to the plant morphology, avoid disconnected corners in the plot and allow comparison with previous measurements (Cerdà and Doerr, 2005). Each plot had a drainage port connected to a 30 L-container through a 40-mm diameter pipe. Each plot had similar surface, although each plot was unique with some differences in size such as can be seeing in Table SP1. Supplementary materials SP1 and SP2 show the size of each plot and their cover in plants, rock fragment and litter cover, plant heights, organic matter and calcium carbonate content (0–2 cm depth) of the soil for each plot.

The sixteen plots were selected on the same slope, but under different canopy cover: *Q. coccifera* (4 plots), *P. lentiscus* (4 plots), *U. parviflorus* (4 plots), and *R. officinalis* (4 plots). The plots were on average 0.38 m² and ranged from 0.33 to 0.45 m² (Supplementary material, SP1). The plots were located underneath the four selected plants and this is why we developed small plots of different size as they were built to adapt to the topography and the plant size. Previous research used larger plots (1-m²) under trees species such as *P. halepensis* and *Q. coccifera* (Cerdà et al., 2017), which is not feasible to apply in shrubland species that use to be smaller than 0.2 m³. The objective was to identify each plot with one species of the Mediterranean shrubland. Larger plots would result in different species present in the same plot and then the hydrological and erosional response would be not determined by one species, rather by multiple species, which was not the objective of this research as our objective is to determine if different plants result in different runoff and sediment yield.

Once the plots were installed, and after each rainfall event, runoff discharge and runoff sediment concentrations were measured, and plot borders, drainage, collectors, pipes and deposits checked. Soil erosion rates and runoff coefficients were calculated for each rainfall event. Events were separated when 6 h without rainfall were recorded. Rainfall data was supplied from the AEMET (Agencia Estatal de Meteorología) meteorological station located 5 km from the study site. Soil and vegetation description and sampling were done in December 2009 before the experiments. Sixty-four (four per plot) soil samples at a depth of 0–2 cm were collected to determine calcium carbonate content, grain size (USDA classification), organic matter and bulk density (see supplementary material). Soil organic matter was determined by the Walkley-Black method (Walkley and Black, 1934). Bulk density was measured using the ring method and calcium carbonate with the potassium dichromate method. Grain size was determined using the pipette method (Deshpande and Telang, 1950). Vegetation cover in each plot was measure with a grid of 1 cm² and the data offered as % of cover (Cerdà et al., 2021a, 2021b) and the stems of each plant were measured with a calliper.

Statistical analyses were performed in order to evaluate the difference in runoff and erosion between *Q. coccifera*, *P. lentiscus*, *U. parviflorus*, and *R. officinalis*. Descriptive statistical tests, such as mean and standard error were calculated. Differences in soil cover and soil properties, runoff, runoff coefficient, sediment concentration and erosion rates variables were analysed using a generalised linear mixed model with binomial

distribution. The plant stand type (*Q. coccifera*, *P. lentiscus*, *U. parviflorus*, and *R. officinalis*) was fitted as the fixed effect, the plot and rainy event as the random effects. The ANOVA results for the generalised linear mixed model are also summarized. Then, an analysis of similarities (ANOSIM), described by Clarke (1993), was carried out for the soil properties and covers, runoff, sediment concentration, sediment yield and soil erosion and the multivariate resemblances were analysed, according to the species factor (four species). Secondly, the non-metric Multi-Dimensional Scaling (MDS) and the Kruskal stress formula (minimum stress: 0.01) were applied to soil properties and covers, runoff, sediment concentration, sediment yield and soil erosion to evaluate the similarity level in the individual cases of the dataset. All statistical analyses were run in R 3.2.0 (R Core Team, 2015). The generalised linear mixed models used the lme4 package (Bates et al., 2015). For the ANOSIM and MDS analyses we used the software PRIMER V7®.

3. Results

3.1. Soil and vegetation cover

Q. coccifera, *P. lentiscus*, *Q. coccifera*, and *R. officinalis* are the main vegetation cover species in the Sierra de Enguera shrubland. *U. parviflorus* and *R. officinalis* recover after fires due to their large seed bank, meanwhile *Q. coccifera* and *P. lentiscus* sprout. All the four plants cover more than 90% of the soil surface. The litter cover that ranges from 22% in *U. parviflorus* plot up to 45% in *Q. coccifera* (Supplementary material SP1). After 11 years of post-fire recovery, shrubland vegetation was dense enough to cover the soil.

The soil has shown a bulk density 1 g cm^{-3} (SE 0.01) in all the plots, and rock fragments cover 20% of the plots, with no significant differences between plots. Organic matter is slightly larger in *Q. coccifera* (7.07%; SE, 0.4) and *P. lentiscus* (6.68%; SE, 0.54) than in *U. parviflorus* (5.37%; SE 0.62) and *R. officinalis* (5.39%; SE 0.74). Calcium carbonate content shows similar values between plots and plant covers, and these range from 57.33% in *P. lentiscus*, to 60.66% in *R. officinalis* (Supplementary material, SP2).

The soils contain around 2% cobbles and 6% gravel, and the soil texture size is sandy clay loam with 45% sand, 20% silt and 35% clay, with almost no differences between plots and plant covers. The electrical conductivity of the soils also shows no variability within plots and ranged from $147.55 \mu\text{S cm}^{-1}$ in *P. lentiscus* and $155.5 \mu\text{S cm}^{-1}$ in *R. officinalis*.

All the described physicochemical soil properties and soil cover variables generate two clear clusters: larger shrubs/small trees with broader leaves and resprouting reproductive strategy (*Q. coccifera* and *P. lentiscus*) versus small needle-like leaved shrubs with seeder reproductive strategy (*U. parviflorus* and *R. officinalis*) when analyzing species factor using a multidimensional scaling routine (Fig. 1). These two groups were statistically different according to the ANOSIM routine (Global $R = 0.32$, statistical significance = 0.1%).

3.2. Rainfall

During the five hydrological years (from October 1st 2010 until September 30th 2014) the total rainfall was 544.2 mm on average per year, ranging from 422.4 and 593.7 mm for 2014 and 2012, respectively. Fig. 2 shows the daily rainfall and the event rainfall, and that extreme rainfall events are rare. From the 150 rainfall events, 48 were with less than 5 mm of rainfall and amounted to 131 mm of rainfall, less than 5% of the total rainfall. However, the four events with more than 100 mm of rainfall amounted to 535 mm and almost 5% of the total rainfall.

3.3. Runoff

The runoff volume measured after each rainfall event was computed per plot and year, and per the five-year experimental period with the SE data (Supplementary material, SP3). The *U. parviflorus* (126.96 L) and

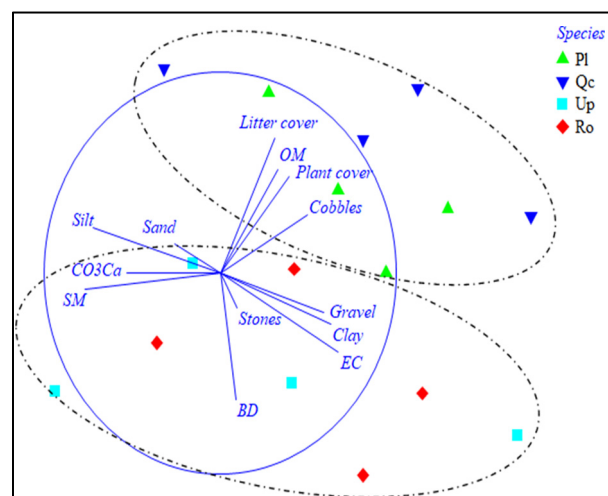


Fig. 1. Multidimensional scaling plot (2D stress 0.18) for all physicochemical soil properties and soil cover variables among species. *Ulex parviflorus* Pourr., (*Up*), *Rosmarinus officinalis* L. (*Ro*), *Quercus coccifera* L. (*Qc*) and *Pistacia lentiscus* L. (*PI*). EC: Electrical conductivity, BD: Bulk density, SM: Soil moisture, OM: Organic matter.

R. officinalis (135.99 L) plots have runoff discharge more than twice as *Q. coccifera* (54.6 L) and *P. lentiscus* (67.31 L) for the five years of research. The total discharge of the whole set of 16 plots was 384.85 L, with some years and plots with high discharge such as plot 4 of *R. officinalis* that had 434.13 L, meanwhile plot 1 of *Q. coccifera*, had 118.7 L.

From the 2709.6 mm of rainfall at the Sierra de Enguera study site from 2010 to 2014 the total runoff produced by the 16 plots was 261.11 mm. The discharge per plot and plant cover was contrasted mainly between *U. parviflorus* (363.46 mm) and *R. officinalis* (375.01 mm) compared to *Q. coccifera* (131.45 mm) and *P. lentiscus* (174.03 mm) for the five years of research. At the yearly scale, the largest runoff discharge was found on plot 4 in 2013 for *R. officinalis* (141 L) and the lowest discharge for *P. lentiscus* (1.62 L) in 2010.

The runoff was calculated as a percentage of the rainfall to give a dimensionless parameter due to the differences in the size of the plots. Runoff reached higher values on *U. parviflorus* (13.41%) and *R. officinalis* (13.84%) than on *Q. coccifera* (4.87%) and *P. lentiscus* (6.42%). Annual runoff values lower than 1 mm were only found in *Q. coccifera* and *P. lentiscus* plots in 2010, meanwhile values higher than 20% were only found in *U. parviflorus* and *R. officinalis* in 2011, 2012 and 2013 (Table 1). The average runoff for the whole research

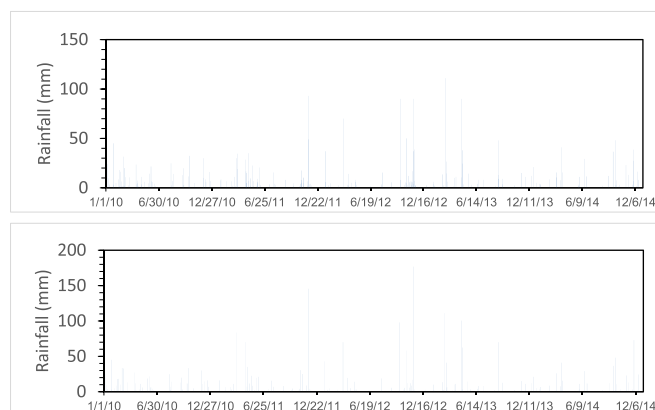


Fig. 2. From up-down, daily rainfall (mm) and Event rainfall (mm) from October 1st 2010 to October 2nd 2014.

Table 1
Runoff discharge (%) per plot, average and total for each plant cover.

%	Runoff						
Species	Plot	2010	2011	2012	2013	2014	Total
PI	1	1.81	10.02	11.57	11.28	1.55	7.66
PI	2	3.70	9.09	13.79	8.41	1.77	6.79
PI	3	0.29	7.31	18.95	4.69	1.68	5.85
PI	4	1.86	8.58	12.35	5.73	1.20	5.39
Average		1.91	8.75	14.16	7.53	1.55	6.42
SE		0.70	0.57	1.66	1.48	0.13	0.51
Qc	1	0.53	7.15	9.26	7.04	1.17	4.38
Qc	2	2.14	5.94	11.16	6.66	1.36	5.01
Qc	3	3.44	7.80	9.45	7.36	1.75	5.49
Qc	4	0.91	6.85	10.99	7.66	2.08	4.60
Average		1.75	6.94	10.21	7.18	1.59	4.87
SE		0.66	0.39	0.50	0.21	0.20	0.24
Up	1	3.16	18.25	17.69	17.86	6.74	13.64
Up	2	3.65	17.70	19.97	11.61	7.39	11.98
Up	3	7.82	16.00	22.61	16.81	5.68	13.25
Up	4	4.20	18.87	18.74	19.60	6.51	14.79
Average		4.71	17.71	19.75	16.47	6.58	13.41
SE		1.06	0.62	1.06	1.72	0.35	0.58
Ro	1	2.59	14.32	20.60	15.27	6.70	10.80
Ro	2	5.50	20.19	21.06	22.41	5.88	15.65
Ro	3	4.47	16.86	21.11	19.54	7.26	12.89
Ro	4	1.60	19.98	19.84	25.16	9.70	16.02
Average		3.54	17.84	20.65	20.60	7.39	13.84
SE		0.88	1.40	0.30	2.11	0.82	1.23

PI, *Pistacia lentiscus* L.; Qc, *Quercus coccifera* L.; Up, *Ulex parviflorus* Pourr.; and Ro, *Rosmarinus officinalis* L.

period and all the plots was 9.64% of the rainfall. According to the multidimensional scaling routine, the runoff was always different between two distinct groups' plots (Qc "Q. coccifera" and PI "P. lentiscus" versus Ro "U. parviflorus" and Up "U. parviflorus") during the 5 years of the experiment (Fig. 3). The ANOSIM routine statistically clustered three different groups (Global $R = 0.72$, statistical significance = 0.1%) such as Fig. 3 shows.

3.4. Sediment concentration

Yearly, sediment concentration in runoff ranged from 7.42 g L^{-1} in 2013 in plot 3 of *R. officinalis*, meanwhile the lowest sediment concentration was registered in 2010 plot 3 of *Q. coccifera* for the five-year research period, the sediment concentration ranged from *U. parviflorus* (4.31 g L^{-1}) and *R. officinalis* (4.88 g L^{-1}) to *Q. coccifera* (4.33 g L^{-1}) and *P. lentiscus* (3.91 g L^{-1}). The average sediment concentration for the whole research period and all the plots was 4.35 g L^{-1} (Table 2). According to the multidimensional scaling routine, the runoff sediment concentration was different when comparing the Ro with Qc, PI and Up, being the latter species grouped on the same cluster (ANOSIM routine, Global $R = 0.43$, statistical significance = 0.5%) (Fig. 3). Peak sediment concentrations were similar for all the 16 plots, with the highest values on August 28th, 2013 when a summer thunderstorm reached 70 mm in one day and the highest sediment concentrations were measured (Table 3).

3.5. Sediment yield and soil erosion

The sediment yield measures the amount of sediment delivered from the plots. For the five years studied *Q. coccifera* show the lowest mean value with 235 g (SE, 7.26), and *R. officinalis* with 663.9 g (SE, 37.46). This loss from *R. officinalis* plots is almost three times more soil lost than *Q. coccifera* plots during 5 years of measurements, demonstrating that the plant species are a key factor that determine the soil losses. Yearly there are large differences with low values in 2010 (5.4–22.74 g) or 2014 (4.64 and 26.02 g). However, other years show higher soil losses such as 2013 with values ranging from 81.54 g in *P. lentiscus*

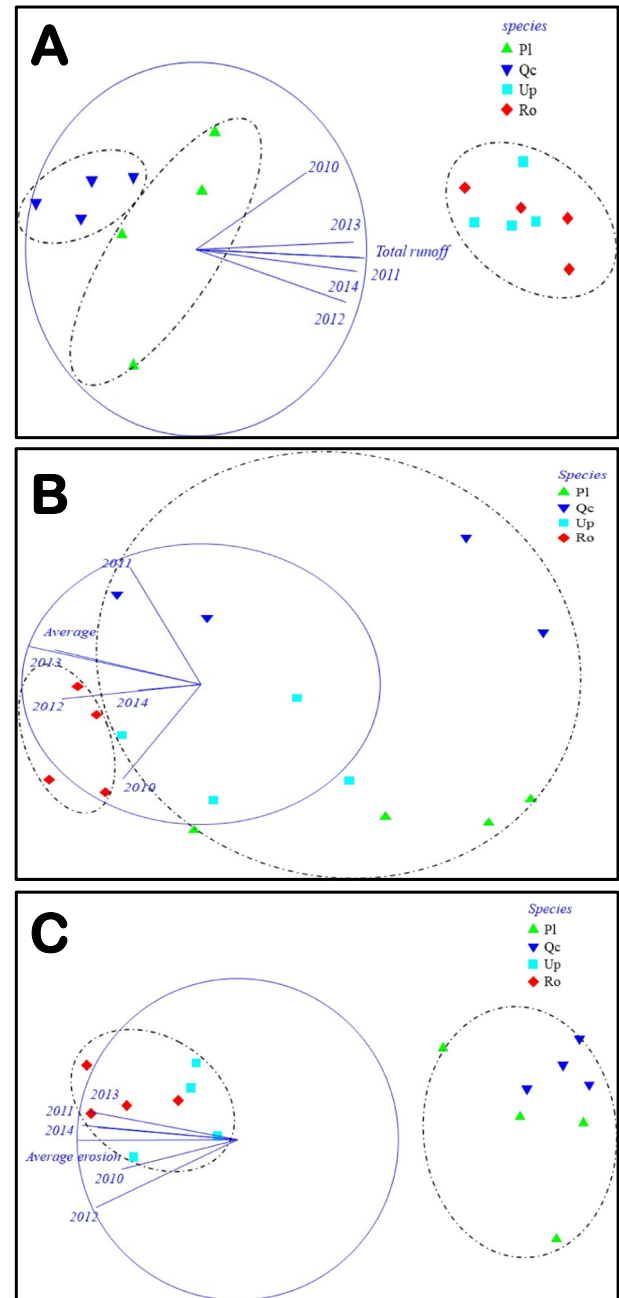


Fig. 3. Multidimensional scaling plot (2D stress 0.02) for (a) all runoff surveyed years (2010–2014 and total runoff), (b) sediment concentration surveyed years (2010–2014 and average runoff sediment concentration), and (c) multidimensional scaling plot (2D stress 0.03) for all runoff surveyed years (2010–2014 and average erosion) and among species. *Ulex parviflorus* Pourr., (Up), *Rosmarinus officinalis* L. (Ro), *Quercus coccifera* L. (Qc) and *Pistacia lentiscus* L. (PI).

and 274.19 g in *R. officinalis* (Table 2). Soil erosion at the four plots ($\times 4$ repetitions) shown contrasting responses from year to year upon the rainfall characteristics. The lowest soil erosion rates were found in *P. lentiscus* ($0.04 \text{ Mg ha}^{-1} \text{ y}^{-1}$) in 2010 and the highest with $9.86 \text{ Mg ha}^{-1} \text{ y}^{-1}$ with *R. officinalis* in 2013 (Table 4). After 5 years of measurements, the rank of soil erosion from lower to higher was: *Q. coccifera* ($1.36 \text{ Mg ha}^{-1} \text{ y}^{-1}$), *P. lentiscus* ($1.36 \text{ Mg ha}^{-1} \text{ y}^{-1}$), *U. parviflorus* ($3.17 \text{ Mg ha}^{-1} \text{ y}^{-1}$) and *R. officinalis* ($3.85 \text{ Mg ha}^{-1} \text{ y}^{-1}$) (Table 4). According to the multidimensional scaling routine, the sediment yield and soil erosion were always different among species (Qc and PI versus Ro and Up) plots during the 5 years of the experiment

Table 2
Sediment yield (g L^{-1}) per plot, average and total for each plant cover.

Sediment yield (gr)							
Species	Plot	2010	2011	2012	2013	2014	Average
PI	1	9.93	80.04	87.91	135.86	6.06	319.80
PI	2	15.36	60.27	99.68	78.86	4.27	258.44
PI	3	1.51	56.79	145.18	47.21	5.14	255.83
PI	4	6.61	59.15	86.33	64.23	3.10	219.43
Average		8.35	64.06	104.78	81.54	4.64	263.38
SE		2.91	5.38	13.79	19.23	0.63	20.81
Qc	1	1.75	67.16	67.60	94.39	3.82	234.71
Qc	2	8.35	60.89	87.52	95.54	3.81	256.12
Qc	3	9.09	71.85	71.13	70.99	4.16	227.21
Qc	4	2.50	67.26	74.95	70.63	8.40	223.74
Average		5.42	66.79	75.30	82.89	5.05	235.44
SE		1.92	2.25	4.34	6.98	1.12	7.26
Up	1	13.23	166.98	120.95	207.61	23.69	532.48
Up	2	18.09	151.43	174.18	128.31	28.59	500.59
Up	3	38.07	134.09	203.88	229.86	15.39	621.30
Up	4	21.55	156.21	136.14	200.14	20.61	534.65
Average		22.74	152.18	158.79	191.48	22.07	547.25
SE		5.39	6.85	18.74	21.98	2.77	25.88
Ro	1	12.87	156.85	181.26	185.60	26.12	562.71
Ro	2	26.30	183.69	190.88	296.72	14.95	712.54
Ro	3	20.98	149.19	174.51	280.44	27.22	652.34
Ro	4	8.77	163.88	185.47	333.98	35.80	727.89
Average		17.23	163.40	183.03	274.19	26.02	663.87
SE		3.95	7.40	3.46	31.58	4.28	37.46

PI, *Pistacia lentiscus* L.; Qc, *Quercus coccifera* L.; Up, *Ulex parviflorus* Pourr.; and Ro, *Rosmarinus officinalis* L.

(Fig. 3). The ANOSIM routine statistically clustered two different groups (Global $R = 0.73$, statistical significance = 0.1%).

4. Discussion

Runoff and soil erosion evaluation under different soil cover and plant species is of paramount importance for properly managing wildfire-affected ecosystems and plan the restoration programs. Bautista et al. (2007) modelled runoff generation based on the spatial distribution of the plants in their study site and demonstrated the importance of the plant cover and plant distribution on soil erosion rates. An update of the knowledge is found in Mayor et al. (2009). Here, through a long-term in situ experimental approach under natural rainfall for five years, we have contributed basic information to advance our understanding of how plant species can affect runoff generation and sediment delivery after wildfire. Our research contributes to the scientific community with a long-term dataset based on natural rainfall (Fig. 4).

4.1. Shrubs as a water and soil conservation plant cover

Shrubs preserve the soil from being lost and induce rainfall to infiltrate in contrast to many soils recently affected by fires. The sixteen plots monitored for five years at the Sierra de Enguera site show that the runoff discharge from the shrubland is low. Out of the 2720 mm of

Table 3
Peak sediment concentration in the runoff (g L^{-1}) per plot and average for each plant cover.

Plant/plot	1	2	3	4	Average
PI	10.32	12.43	10.32	15.30	12.09
Qc	12.43	14.21	12.23	12.87	12.94
Up	11.13	14.21	14.87	11.32	12.88
Ro	11.76	13.76	13.23	13.65	13.10

PI, *Pistacia lentiscus* L.; Qc, *Quercus coccifera* L.; Up, *Ulex parviflorus* Pourr.; and Ro, *Rosmarinus officinalis* L.

Table 4
Soil erosion ($\text{Mg ha}^{-1} \text{y}^{-1}$) per plot, average and total for each plant cover.

Mg ha ⁻¹ y ⁻¹	Soil erosion						
Species	Plot	2010	2011	2012	2013	2014	Average
PI	1	0.29	2.32	2.55	3.94	0.18	1.85
PI	2	0.45	1.75	2.89	2.29	0.12	1.50
PI	3	0.04	1.65	4.21	1.37	0.15	1.48
PI	4	0.19	1.71	2.50	1.86	0.09	1.27
Average		0.24	1.86	3.04	2.36	0.13	1.53
SE		0.08	0.16	0.40	0.56	0.02	0.12
Qc	1	0.05	1.95	1.96	2.74	0.11	1.36
Qc	2	0.24	1.77	2.54	2.77	0.11	1.48
Qc	3	0.26	2.08	2.06	2.06	0.12	1.32
Qc	4	0.07	1.95	2.17	2.05	0.24	1.30
Average		0.16	1.94	2.18	2.40	0.15	1.36
SE		0.06	0.07	0.13	0.20	0.03	0.04
Up	1	0.38	4.84	3.51	6.02	0.69	3.09
Up	2	0.52	4.39	5.05	3.72	0.83	2.90
Up	3	1.10	3.89	5.91	6.66	0.45	3.60
Up	4	0.62	4.53	3.95	5.80	0.60	3.10
Average		0.66	4.41	4.60	5.55	0.64	3.17
SE		0.16	0.20	0.54	0.64	0.08	0.15
Ro	1	0.37	4.55	5.25	5.38	0.76	3.26
Ro	2	0.76	5.32	5.53	8.60	0.43	4.13
Ro	3	0.61	4.32	5.06	8.13	0.79	3.78
Ro	4	0.25	4.75	5.38	9.68	1.04	4.22
Average		0.50	4.74	5.31	7.95	0.75	3.85
SE		0.11	0.21	0.10	0.92	0.12	0.22

PI, *Pistacia lentiscus* L.; Qc, *Quercus coccifera* L.; Up, *Ulex parviflorus* Pourr.; and Ro, *Rosmarinus officinalis* L.

rainfall during the study period, 261 mm were yield as runoff although the plots covered a small area, which use to induce higher runoff discharge (Martinez et al., 2017). Runoff was 9.6% of the rainfall. Similar results were found related to the soil erosion rate that was registered as $2.48 \text{ Mg ha}^{-1} \text{y}^{-1}$ for five years and 16 plots' average. In comparison to other land uses and land managements, shrubland is a sustainable conservation plant cover. As an example of other land uses under similar climatic conditions and plot scale the data show that agriculture land contributes higher soil losses than shrubland such as was estimated recently by Ricci et al. (2020) in Plugia with values of $5 \text{ Mg ha}^{-1} \text{y}^{-1}$. In Morocco, Meliho et al. (2019) measured using fallout ^{137}Cs soil loss values of $8.5 \text{ Mg ha}^{-1} \text{y}^{-1}$ in cereal crops, $6 \text{ Mg ha}^{-1} \text{y}^{-1}$ in groves and orchards, and in the forest $4.2 \text{ Mg ha}^{-1} \text{y}^{-1}$. Other recent research by Rubio-Delgado et al. (2019) based on topographic surveys, measured in the Parapuños experimental site values of 7.4, 18.5 and $29.1 \text{ Mg ha}^{-1} \text{y}^{-1}$ during different historic times in the wooded rangelands in Extremadura. Bayat et al. (2019) measured in Valencia values of $50 \text{ Mg ha}^{-1} \text{y}^{-1}$ in persimmon plantations by means of the ISUM method and Rodrigo-Comino et al. (2020) using the same method measured soil erosion rates of $31.9 \text{ Mg ha}^{-1} \text{y}^{-1}$ for herbicide-treated soils and $52.6 \text{ Mg ha}^{-1} \text{y}^{-1}$ for the tilled ones. Rellini et al. (2019) applied the RUSLE soil erosion model in the Liguria region and found values ranging from 0 to $225 \text{ Mg ha}^{-1} \text{y}^{-1}$ and a mean value of $9 \text{ Mg ha}^{-1} \text{y}^{-1}$. Barrena-González et al. (2020) estimated $45.7 \text{ Mg ha}^{-1} \text{y}^{-1}$ with the USLE erosion model and $17.4 \text{ Mg ha}^{-1} \text{y}^{-1}$ with the ISUM method in vineyards. Rainfall simulation experiments also contributed recently with information about the high erosion rates in agriculture land of vineyards (Rodrigo Comino et al., 2018), Persimmon plantations (Cerdà et al., 2021a) and on experimental plots (Cerdà et al., 2021b, 2021c) where values of soil losses are one or two order of magnitude higher than in the shrublands. Definitely, shrubland are a protective cover that must be promoted in the Mediterranean Type-Ecosystems such as already other colleagues found (Bochet et al., 2006) under diverse plant types from semiarid ecosystems. Johansen et al. (2001) found in fire affected land the role of plants to control soil and water losses and Li et al. (2000) how they determine the spatial distribution of the soil and water losses.

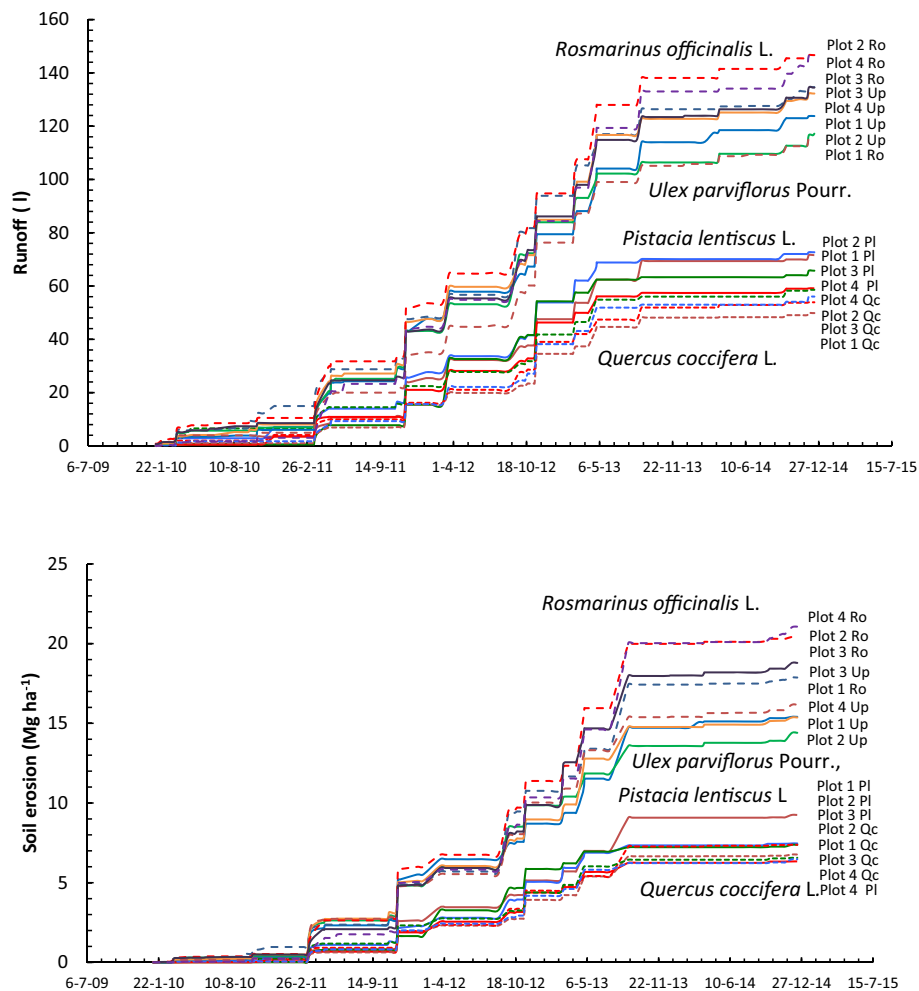


Fig. 4. Accumulated runoff (l) and soil erosion (Mg ha⁻¹) per plot during the five years (2010–2014) for the four studied species: *Ulex parviflorus* Pourr., (Up), *Rosmarinus officinalis* L. (Ro), *Quercus coccifera* L. (Qc) and *Pistacia lentiscus* L.

On the rangelands and forest lands of the Mediterranean, previous measurements have found that soil erosion is low. At the Sierra de Enguera site, Cerdà et al. (2017) measured runoff rates of 8.31 and 1.39% runoff for *P. halepensis* and *Q. ilex*, respectively. The soil erosion rates after 5 years of measurements were on average 2.57 and 0.26 Mg ha⁻¹ y⁻¹. Vega et al. (2005) measured 0.05 Mg ha⁻¹ y⁻¹ in gorse shrubland in Galicia. Gimeno-García et al. (2007) measured 0.08 Mg ha⁻¹ y⁻¹ in shrubland shrublands in Valencia. Similar findings were recorded by Whicker et al. (2002), Breshears et al. (2003) and García-Estringana et al. (2010a). However, here we have an effect determined by the impact of the length of the plot and the connectivity of the flows that are affected by the scale of the plot (Urgeghe and Bautista, 2015; Keesstra et al., 2018). The low soil erosion rates found in shrublands increase as a result of disturbance by forest fires (Johansen et al., 2001), or human-induced activities such as trail and dirt road construction and use (Salesa and Cerdà, 2019), but under natural conditions, shrublands is a sustainable vegetation cover from the point of view of soil and water conservation. Casermeiro et al. (2004) confirmed with simulated rainfall experiments the positive effect of dwarf shrubs in controlling soil losses in the Central Iberian Peninsula. Although the data are collected by different methods and geographical regions of the Mediterranean, shrubland is considered a protective cover of the soil. We confirm the low erosion rates found in Sierra de Enguera, and we contribute with data that inform which shrubs species are the most successful to control the soil and water losses.

4.2. Runoff and soil erosion temporal changes

Our detailed measurements of the runoff and sediment delivery in Sierra de Enguera contribute with information about the temporal changes in soil and water losses. The sediment and water yield in the shrubland ecosystem are concentrated in a few rainfall events. Runoff yield (l) accounted for 79.37% of the total runoff in 10 rainfall events that contributed 1005 mm out of the 2709.6 mm of rainfall. This means that 37.1% of the rainfall yielded nearly 80% of the runoff. Most of the rainfall events do not contribute runoff due to the low volume or intensity of the rainfall. The rank of rainfall events ordered by runoff and soil erosion is shown in the supplementary material (SP6). Data shows that some rainfall events that contribute high runoff discharge were not relevant for the soil losses. The March 2nd, 2012 event registered 12.57% of the total runoff generated in 5 years. However, this event was not even ranked within the 10 soil erosion events. This was due to the fact that this event was mainly snow and thus infiltration was very high and runoff negligible. Another example is the case of the highest soil losses triggered by the event that was measured on August 29th 2013, when a 70 mm rainfall event contributed 18.14% of the soil losses in five years, meanwhile the runoff discharge was not in the 10 most important runoff events (see supplementary material SP6).

Soil erosion was concentrated in fewer events than runoff. Five rainfall events triggered 71.73% of the sediment yield in five years. For the runoff, five rainfall events contributed 52.57% of the runoff yield. The

concentration of soil erosion in few rainfall events was also reported by other authors such as Edwards and Owens (1991) as they found that within 4000 rainfall events, 66% of the soil losses could be attributed to five events over 28 years. González-Hidalgo et al. (2007) reported that the three-year highest daily erosive events contributed more than 50% of the annual soil eroded in the western Mediterranean areas. In USA, based on the USLE database, the most extensive daily soil erosion dataset, Gonzalez-Hidalgo et al. (2012) found soil erosion is highly time compressed and is independent of the climatic conditions. Based on this finding, Gonzalez-Hidalgo et al. (2012) concluded that the number of erosive events is the key information needed to characterize soil erosion in a region, and not the number of years of measurements, or the rainfall events with low intensity. What Larson et al. (1997) characterized as severe storms is the key to understand soil erosion processes, to model them and to plan soil erosion control measures. At the Sierra de Enguera research site, we found that after 213 rainfall days in 5 years and 150 runoff events, 10 of the events accounted for 91.7% of the soil losses.

4.3. The plant effect on soil and water losses. Factors

The shrub cover studied at the Sierra de Enguera study shown 70% of the ground covered along the 12 ha of the study site, and with values above 90% in the patches where plants concentrate. As the vegetation cover showed almost no inter-annual changes, rainfall was the key factor that controlled the soil and water losses. The two seeder species (Ro and Uf) contributed with higher soil losses (605 g in average for five years) meanwhile the other two species (Qc and Pl) resulted in lower soil losses (249 g) (Fig. 5). We measured differences in soil erosion and runoff generation among four species, but few differences were found in our measurements on the soil and plants characteristics, except for the differences in litter cover (see supplementary material), that is, *U. parviflorus* (22%) and *R. officinalis* (24%) have less litter cover than *Q. coccifera* (45%) and *P. lentiscus* (33%). Litter has been found to be a key factor in soil erosion and runoff generation in agricultural land (Liu et al., 2017) and forest land (Wang et al., 2017). The influence of litter is definitive as protect the soil from the raindrop impact and promote the biota. In general, plants controls the hydrological cycle (Keesstra et al., 2012).

Vegetation structure affects the dynamics of surface runoff and groundwater discharge, mostly through growth habit and ecophysiological performance of dominant species. The specific soil-plant-atmosphere water continuum which is established by a particular species depends on numerous plant traits that contribute to successful stand recolonization after the fire event and development of plant cover with soil-recovering capacity (Passioura, 1982). Four species native to Mediterranean shrubland assessed in this study show some similar traits in terms of growth habit (shrubs) and evergreen, hard, sclerophyllous

leaves, which reflect both structural and functional adaptations, i.e. a balance between carbon input and water loss (Specht et al., 2012). However, maximum plant height and leaf shape differ among the four studied species: *Rosmarinus officinalis* (43 cm) and *Ulex parviflorus* (54 cm) are shrubs about 0.5 m tall, with stem diameters of a 9.8 mm and 7.8 mm and needle-like leaves, whereas *Quercus coccifera* (82 cm) and *Pistacia lentiscus* (84) reached 1 m tall at the study sites after a decade without fires, and the stems were 34.5 mm and 31.3 mm in diameter with wide leaves. All of these differences affect not only total plant biomass and biomass of leaves (relevant for plant cover and litter cover, and also for the ability of xylem to conduct water to the leaves), but also transpiration and intercepting leaf area which are the most relevant for plant water balance and plant-soil relations (Larcher, 2003).

Our research shows statistically significant differences in runoff and soil erosion from seeders and sprouters. Different soil properties can determine the differences between plant species response to rainfall. Key factors found in the scientific literature are the soil structure and porosity. Within these factors, macroporosity can affect the runoff discharge and thus the soil erosion. This is the conclusion of a long-term research project by Zhang et al. (2007) after 24 years with conservation tillage where soil structure and macroporosity changes the fate of the runoff discharge. Abu-Hamdeh et al. (2018) also found that different tillage systems affect the runoff generation. Future research projects at the Mediterranean shrubland should focus on soil properties and plant characteristics to go further to understand the relationship between soil, water, erosion and plants. Another potential factor that can determine the hydrological response between plant species is their canopy cover. Wang et al. (2005) found that the rainfall interception ranged from 6.9% to 11.7% after 40 years of restoration for *Artemisia ordorsica* and *Caragana korshinskii*, respectively, at the Shapotou Desert Experimental Research Station in China. Wood et al. (1998) measured the rainfall interception of 10 common species in the Chihuahuan Desert and found contrasting responses such as was found by Tromble (1983) for twelve tarbush species from southern New Mexico. Measurements of rainfall interception by shrubs are rare as they need special equipment (Belmonte-Serrato and Romero-Diaz, 1998; García-Estringana et al., 2010b). More information is available for the trees such *Quercus ilex*, but similar interception rates were measured, such as the 22% interception of the gross rainfall reported by David et al. (2006). Navar and Bryan (1990) measured 27.32% of rainfall interception in individual shrubs of *Diospyros texana*, *Acacia farnesiana* and *Prosopis laevigata* in northeastern Mexico. Future research in the Mediterranean shrubs should improve the knowledge on interception. Moreover, soil water repellency must be studied to understand the impact that plants have on the hydrological processes. Seaton et al. (2019) found that 92% of the soils in Wales during the summer season shown water repellent responses. This shows how relevant is the water repellency. Plants are the source of the soil organic matter via the litter, and they can transfer to the soil substances that are highly water repellent. Water repellency determines the runoff discharge and water infiltration pattern at the pedon scale. This is why different plant cover results in different water-repellent responses such as Zema et al. (2021) found in Cuenca (Spain) pine forest and Benito et al. (2019) found in Galicia. Usually, a higher water repellency degree in the soil results in more runoff. More runoff is also a general trend in fire affected land, although not always is due to higher water repellency as fire can reduce the water repellent substances via volatilization (Robichaud and Hungerford, 2000).

In our study case in Sierra de Enguera, a fire affected the land in the last century, fire is found a key factor that will determine the runoff discharge and the soil water repellency after fires. Zvirzdin et al. (2017) found this relevance of water repellency in the piñon-juniper woodland in USA and Plaza-Álvarez et al. (2018) in central Iberian Peninsula. Our results on soil erosion and runoff inform about how plants affect the runoff discharge, and this informs that the water repellency can be behind this contrasted behavior and should be researched to fully understand the hydrological and erosional response of soils. The

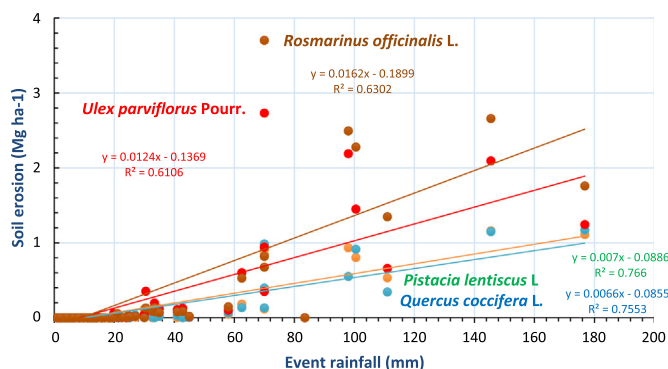


Fig. 5. Relationship between the total rainfall of an event and the soil erosion (Mg ha^{-1}) mean value per plant species (four plots average) during the period 2010–2014 for four studied species: *Ulex parviflorus* Pourr., (Up), *Rosmarinus officinalis* L. (Ro), *Quercus coccifera* L. (Qc) and *Pistacia lentiscus* L. (Pl).

water repellency and then any surface wash during rain events is dependent on the plant species as some species promote high runoff discharge and others do not. Water repellency is not a new research issue (Adams et al., 1970), neither is the influence of plants on soil water repellency (Savage et al., 1972). However, how different plants result in different runoff discharges and soil erosion rates as a result of water repellency should be investigated in greater detail to determine why some plants contribute to greater runoff and soil losses than others.

A fourth topic that should be researched in the future investigations is the root system. Sprouters shown lower runoff and soil losses probably due as their roots survive to the forest fires and macropore flow is already established. The seeders should create after each forest fire a new root system and this is very relevant for the trade-off between vegetation and soil erosion and runoff generation such as Wu et al. (2020) found in a recent meta-analysis. Litter was found as a key factor to control the soil erosion in shrublands as sprouters soils are better covered than in the seeders, which are also smaller in height. Future research should focus on the effects of microporosity, interception, water repellency and roots system to better understand the runoff generation on Mediterranean shrublands.

The limitation of this research is due to the constraints of all the investigations developed in the field and on plots. The use of closed plots avoids the transfer of sediments along the slope, and then promotes the exhaustion of the sediments available within the plots, and this is why our research was restricted to 5 years, to avoid the effect of the fenced plots. We did not register any reduction in the sediment concentration along the five-year research period. Another limitation of our study is the small size of the plots. In this research, the plots smaller than 1 m² were necessary as the objective was to isolate each plant species in the field. Previous research done in the laboratory or under simulated rainfall experiments contributed with some information (García-Estringana et al., 2010a, 2010b).

5. Conclusions

Vegetation controls the hydrological and erosional cycle and this paper contributes with information about the role of individual plants of different species on runoff and sediment delivery. Five years of rainfall, runoff and sediment measurements in a Mediterranean shrubland demonstrates the protective role of shrubs plants. Within the four shrubs tested, the seeders (*U. parviflorus* and *R. officinalis*) contribute with 2.4 times more runoff than the sprouters (*Q. coccifera* and *P. lentiscus*). The sediment concentration was 1.2 lower in the sprouters and the soil erosion rates were 2.8 times higher in the seeders. Their response to the rainfall was different (sprouters more protective) but in general the shrubs behave as a very protective plant cover. After wildfires, when the runoff and sediment delivery is enhanced, we suggest the use of sprouters within the restoration and rehabilitation programs. The data collected demonstrated also that the soil and water loss take place in few events. 80% of the runoff was generated with only 37% of the rainfall. Ten rainfall events out of 150 produced 92% of the soil erosion in the Sierra de Enguera shrubland. Another relevant finding is that the events with higher soil loss are not the ones with the higher amount of runoff generated and open a new scientific research topic within the shrubland hydrology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2021.149218>.

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